

Process Scheduling - Exam Questions and Answers

Question 1

Describe how round-robin, priority, and lottery scheduling work. Provide an example in which round-robin scheduling performs better than the other two.

Question 2

Given the following table, and assuming Shortest Job First (non-preemptive), show how the CPU slots are allocated to processes and compute the average waiting time. All burst times are expressed in number of required CPU slots.

Process	Arrival time	Burst time
P1	0	10
P2	1	3
P3	3	5
P4	11	2
P5	12	3

Question 3

A variation of the round-robin scheduler is the regressive round-robin scheduler. This scheduler assigns each process a time quantum and a priority. The initial value of a time quantum is 50 milliseconds. However, every time a process has been allocated the CPU and uses its entire time quantum, that is, does not block for I/O, 10 milliseconds is added to its time quantum, and its priority level is boosted. The time quantum for a process can be increased to a maximum of 100 milliseconds. When a process blocks before using its entire time quantum, its time quantum is reduced by 5 milliseconds, but its priority remains the same.

What type of process, CPU-bound or I/O-bound, does the regressive round-robin scheduler favor? Explain.

Question 4

Consider the following set of processes, with the length of the CPU burst time given in milliseconds:

Process	Burst time	Priority
P1	2	2
P2	1	1
P3	8	4
P4	4	2
P5	5	3

The processes are assumed to have arrived in the order P1, P2, P3, P4, P5, all at time 0.

1. Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, non-preemptive priority (a smaller priority number implies a higher priority), and RR (quantum = 1).
2. What is the turnaround time of each process for each of the scheduling algorithms in part 1?
3. What is the waiting time of each process for each of these scheduling algorithms?
4. Which of the algorithms results in the minimum average waiting time, over all processes?